

24th International Conference on Modelling and Applied Simulation (MAS 2025), held within the 22nd International Multidisciplinary Modeling & Simulation Multiconference (I3M 2025)

Sports analytics optimizing injury prevention and player availability in professional soccer

Yashwanth Suresh^a, Roni Avakian^a, Silvia Carpitella^{a,*}, Maryam Tabibzadeh^a, Sepideh Abolghasem^a

^a*Department of Manufacturing Systems Engineering and Management, California State University, Northridge; 18111 Nordhoff St, Northridge, CA 91330, United States*

Abstract

Sports analytics has become an important tool in professional soccer, due to its potential to transfer raw data into valuable insights that support both player performance and team strategy. However, injuries remain a major challenge affecting player availability, team results, and long-term squad management. Existing research identifies some key factors as the main drivers of injury risk, for instance a consecutive exposure to matches, a limited recovery time and unfavorable environmental conditions. Unlike earlier models focused on internal factors, our research also considers external variables like match conditions and opponent difficulty, among others. This paper focuses on collecting, validating and preparing multi-season player data as the basis for a Monte Carlo simulation model producing probabilistic insights on injury prevention and workload management. The results emphasize the importance of data quality in building reliable predictive models that prioritize player health and long-term team performance in the world of professional soccer. Building on this research, the simulation model will offer a foundation for continued investigation into data-driven approaches to injury prevention and player availability in this highly competitive and economically significant industry.

© 2025 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the 22nd International Multidisciplinary Modeling & Simulation Multiconference (I3M).

Keywords: Injury Prediction; Sports Analytics; Data Integrity; Data Collection; Data Preprocessing

1. Introduction and Literature Review

Sports analytics has become essential for understanding and enhancing athletic performance and team dynamics [1]. Coaches and analysts can make informed decisions by understanding certain patterns and trends that are not always visible during play [2]. In the context of modeling and simulation, this insight allows for more accurate scenarios, supporting better planning and strategy development [3]. Specifically, sports analytics has shown strong

* Corresponding author.

E-mail address: silvia.carpitella@csun.edu

potential in predicting injuries in professional soccer [4], offering valuable support for player health and workload management. However, the reliability of current models remains limited due to such ongoing challenges as incomplete data [5], variability of tasks among players [6], and the complex nature of injury mechanisms [7], among others.

Injury prediction in professional soccer presents several well-documented challenges that hinder the development of reliable models. As noted by [8], injury events are relatively rare, resulting in class imbalance that can negatively affect model learning and predictive accuracy. This imbalance means that models are often trained on datasets dominated by non-injury cases, making it harder to recognize patterns leading to injuries. This often causes the model to favor the majority class, reducing its ability to detect true injury risks. Additionally, [9] emphasizes the subjectivity in injury reporting, which complicates data labeling and undermines the long-term validity of predictive efforts. Injury definitions can indeed vary across different surveillance systems within clubs and organizations [10]. This lack of standardization leads to noisy or unreliable labels in the dataset, causing models to learn from inconsistent data and reducing their long-term effectiveness. Another significant barrier has been highlighted by [11] and refers to sub-optimal feature selection, in other terms the use of irrelevant or incomplete variables. This aspect likely limits both model performance and the ability to generalize across teams and seasons, as models may fail to capture the most relevant factors influencing injury risk. Without well-chosen features, predictions may indeed reflect noise rather than meaningful patterns, reducing their reliability in considering contextual factors [12]. Although recent research has increasingly turned to machine learning approaches, including Artificial Intelligence (AI) and Internet of Things (IoT) technologies [13], these models often depend on large, high-quality datasets and can lack transparency, making them difficult to interpret and apply in practice. While the studies herein analyzed collectively outline the key obstacles in the field, practical and widely applicable solutions remain limited, and to the best of the authors' knowledge, few approaches effectively balance predictive accuracy with real-world usability. The present paper aims to address this gap by prioritizing requirements such as high-quality data, rigorous variable selection, and the inclusion of systematic preprocessing, all of which is fundamental when creating a foundation for an effective and scalable simulation-based injury prediction model. This study will serve as a foundation for future research focused on the introduction of a simulation-based method that is transparent, built on structured data, and applicable to real-world scenarios.

This work is structured as follows: following the current section, which outlines the research motivation, objectives, and a review of the relevant literature, Section 2 presents the proposed methodology. Section 3 introduces the simulation framework, describes the expected outcomes and discusses the implications of the expected simulation results. Lastly, Section 4 presents concluding remarks and proposes directions for future research.

2. Proposed Methodological Approach

This paper presents a simulation-based framework for injury prediction using Monte Carlo techniques. The model incorporates two main categories of input: player-specific and contextual variables, derived from five seasons of Liverpool FC first-team data. Instead of generating fixed predictions, the simulation runs thousands of match scenarios, accounting for varying factors such as workload, recovery time, and tactical context to estimate injury risk. Monte Carlo methods have previously been applied in soccer to assess match importance and forecast outcomes, as shown by [14]. This study adapts the technique to model injury risk under dynamic conditions. The dataset is prepared through a structured preprocessing pipeline, including normalization, encoding of categorical variables and class balancing to ensure model integrity.

Figure 1a illustrates the full methodology while Figure 1b outlines the data preparation stages that support accurate scenario generation and risk profiling. As shown in Figure 1a, the methodology follows a step-by-step structure, starting with data collection and progressing through preprocessing, database structuring, simulation, and finally injury risk profiling. At each stage, common data-related issues are flagged, such as missing or inconsistent entries, encoding and normalization issues, and unrealistic simulation results. These checkpoints help identify potential sources of error early in the process, allowing for corrective action before proceeding. This structure also ensures that each phase builds on the quality of the previous step, supporting reliable and interpretable simulation outcomes. Figure 1b breaks down the preprocessing workflow in depth, with the main focus on how raw inputs are transformed into structured, simulation-ready data. It highlights key steps such as manual data verification, injury categorization, class balancing, and the encoding of categorical variables. Such potential error points as incomplete injury tags or schema mismatches are clearly marked to illustrate the risks that can arise when integrating multi-source datasets.

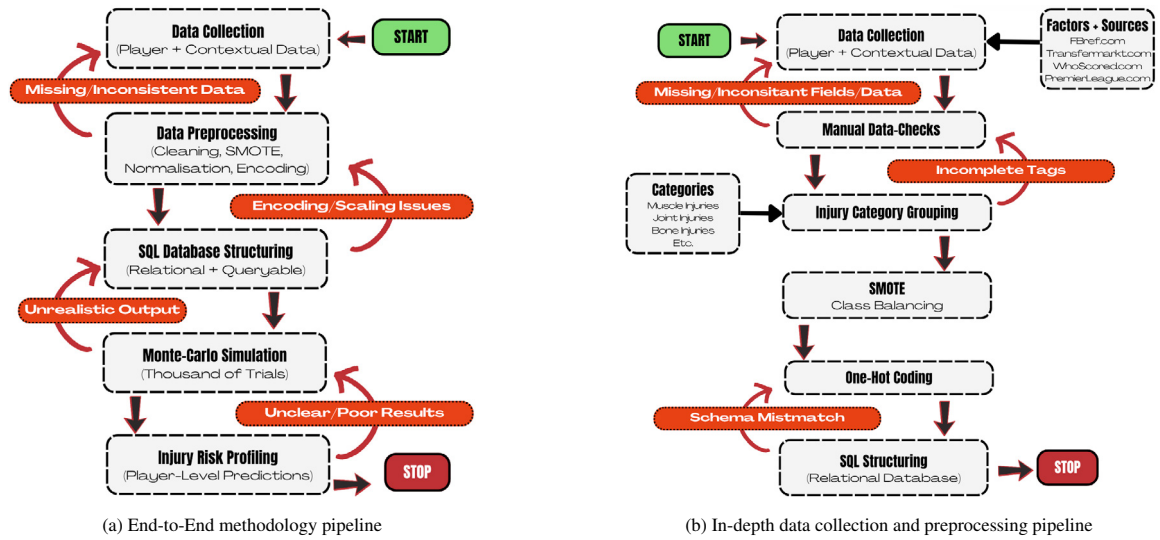


Fig. 1. Overall system pipeline and its preprocessing component

2.1. Data Collection

The data collection phase forms the backbone of the simulation model by defining the structure and scope of the selected variables involved. This stage involves compiling multi-season player data that captures both individual performance and match context. Each data entry acts as an input for the simulation, so the accuracy and consistency of the raw data directly affect the reliability of the risk estimates. By focusing on completeness as well as relevance, the dataset supports robust scenario modeling rather than limiting the simulation to specific cases. This approach enables capturing the variability in player roles, as well as match intensity and recovery periods, all factors that are essential for realistic injury forecasting.

2.1.1. Selection and categorization of predictive factors

As mentioned earlier, the variables used in this study are grouped into two categories, that are player-specific and contextual variables. Table 1 highlights how each variable relates to injury risk in professional soccer, while Tables 2 and 3 provide details about the mentioned categories of variables, where each factor is further classified by data type, i.e., numerical or categorical, and consequently matched with its corresponding distribution or encoding method.

Table 1. Full set of predictive factors and related relevance

No.	Factor	Relevance to injury risk
1	Minutes Played per Game	Reflects physical load and fatigue accumulation, as mentioned by [15]
2	Games Played per Season	Measures cumulative exposure across a competitive period as mentioned by [15]
3	Days Between Matches	Captures recovery time and scheduling congestion as mentioned by [16]
4	Passing Completion	Proxy for pressing intensity and physical exertion as mentioned by [17]
5	Fouls Received per Game	Indicates physical contact and potential for acute impact injuries as mentioned by [15]
6	Player Age	Represents biological recovery rate and injury susceptibility with age as mentioned by [15]
7	Previous Injury History	Personal risk factor leading to higher probability of recurrence as mentioned by [15]
8	Position Played	Linked to role-based physical demands, e.g., defenders vs. attackers as mentioned by [15]
9	Formation Utilized	Tactical context affecting space coverage and movement load as mentioned by [18]
10	Home vs. Away	Contextual stress, e.g., travels, unfamiliar conditions, crowd effect as mentioned by [18]

The variable distributions were determined through visual inspection (e.g., histograms, Q-Q plots), descriptive statistics (e.g., skewness, kurtosis), and alignment with the observed data patterns across five seasons of Liverpool FC match records to ensure realistic sampling during simulation.

Table 2. Player-specific factors

Factor	Type	Distribution / Encoding
Minutes Played per Game	Numerical	Right-skewed, Poisson
Games Played per Season	Numerical	Discrete count, Poisson
Days Between Matches	Numerical	Continuous, Normal
Passing Completion %	Numerical	Log-normal (proxy for pressing intensity)
Fouls Received per Game	Numerical	Discrete, Poisson
Player Age	Numerical	Normal
Previous Injury History	Categorical	Encoded by injury type, e.g., muscle, joint.
Position Played	Categorical	Encoded by role, e.g., defender, midfielder

Table 3. Contextual factors

Factor	Type	Distribution / Encoding
Formation Utilized	Categorical	Encoded by tactical setup (e.g., 4-3-3, 4-4-2)
Home vs. Away	Categorical	Binary (Home = 1, Away = 0)

This structure supports the accomplishment of an accurate preprocessing stage while enabling the generation of context-aware injury scenarios in the Monte Carlo simulation. As it is possible to observe in Table 1, the ten selected factors reflect common contributors such as playing time, recovery gaps, and position-specific demands. These elements have also shown practical value in previous studies. For example, [19] demonstrated that the use of player-specific inputs in injury prediction led to a 13% reduction in overall team injuries. This highlights the importance of selecting variables that are both relevant and clearly interpretable. In the context of the simulation-based framework proposed in this study, such targeted inputs help produce more meaningful and applicable results, supporting decisions that can have a direct impact on player availability and team performance.

2.1.2. Data sources and dataset structure

The dataset for this study was compiled from reputable, publicly available sources selected for their reliability and consistency. Match statistics were gathered from *FBref.com*, injury history from *Transfermarkt.com*, tactical data from *WhoScored.com*, and live injury updates from Premier Injuries. These platforms were chosen for their structured reporting formats and frequent updates. All sources follow transparent data collection methods and provide accessible documentation, supporting both replication and academic standards. Using publicly verifiable data also addresses concerns around security, ethical access, and long-term sustainability of the dataset. Table 4 summarizes the overall dataset, including the team analyzed, the volume of data collected, and the scope of information used to support the simulation.

Table 4. Dataset overview

Category	Relevance to injury risk
Club Covered	Liverpool FC (First-team only)
Seasons Included	2019/2020 to 2023/2024
Players per Season (approx)	~25
Matches Per Player (max)	38 (Premier League Only)
Total Player-Match Entries	~4,750
Factors per Entry	10 Standardized Factors
Total Data Points	~47,500

While public data sources offer transparency and ease of access, they also present limitations that may affect the model's responsiveness and scope. For example, some variables such as tactical setups, player fatigue, and injury events may be affected by reporting delays or inconsistent tagging across platforms. These time lags limit the model's ability to update in real time and restrict the availability of more granular parameters. In addition, other player-specific factors such as, for instance, heart rate, heart rate variability, real-time training loads, or psychological stress

levels, among others, are not publicly available. These elements could improve model precision but they are typically proprietary to teams and medical staff. Although steps like classification and normalization improve structure and usability, the overall framework remains sensitive to the timing and quality of source data. As a result, conservative interpretation and continuous validation are necessary when working with publicly available datasets.

2.2. Data Preprocessing

Before proceeding to the simulation phase, the raw dataset is first prepared using a structured preprocessing workflow. This step enhances data consistency and ensures it aligns with the requirements of the modeling framework. It also addresses common challenges in sports analytics such as the ones previously discussed, i.e., class imbalance, differences in numerical scales as well as the handling of categorical variables.

Being recognized in literature, these challenges need to be addressed through the implementation of appropriate and targeted preprocessing measures. To such an aim, the core preprocessing methods herein proposed include:

- Synthetic Minority Over-sampling Technique (SMOTE), generating examples for minority classes;
- Normalization, aimed at scaling numerical features to ensure consistent ranges across different variables;
- One-hot encoding, aimed at converting categorical variables into a binary format suitable for modeling.

These techniques form the foundation for accurate risk modeling, as shown in Table 5. The additional procedures reported in Table 5, as previously formalized in Figure 1b, are also proposed to enhance data quality. It is important to note that, while not all preprocessing steps directly impact the simulation logic, their integration enhances data integrity and reduce bias, supporting a more robust and reproducible framework across different inputs.

Table 5. Summary of preprocessing techniques before simulation

Technique	Description
Manual Data Checks	Detecting and correcting missing, inconsistent, or duplicate entries
Injury Category Grouping	Categorizing injuries by type and location for uniform trend analysis
SMOTE	Generating synthetic injury cases to correct class imbalance
Normalization	Rescaling continuous factors, e.g., minutes played, to a 0–1 range
One-Hot Encoding	Converting categorical factors into binary format for model input
Relational SQL Structuring	Organizing data into linked tables for efficient access and scalability

2.2.1. Handling missing data

A major challenge in data preparation was handling incomplete or inconsistent injury records, especially from early seasons and international breaks. To improve consistency, all injuries were classified into standardized categories based on anatomical location and type, as shown in Table 6. This approach allowed for clearer trend analysis and ensured uniformity across seasons. Despite the use of standardized categories, class imbalance remained a significant issue due to the low frequency of injury events. To address this, SMOTE was applied. As introduced by [20], SMOTE generates synthetic samples for the minority class through interpolation, rather than simply duplicating existing cases. This increases the diversity of injury cases while reducing the risk of overfitting. Balancing the dataset is essential for future integration of machine learning models, as it ensures more reliable training and better performance on new, unseen data.

2.2.2. Encoding categorical data

To prepare categorical variables for simulation, one-hot encoding was used to convert them into numerical formats that are suitable for this analysis. This procedure avoids potential incorrect assumptions about order while ensuring compatibility with modeling algorithms. More in detail, variables encoded in this way include formation utilized, position played, match location (Home vs. Away), and previous injury history. This transformation allows the model to process qualitative tactical and contextual inputs accurately, without introducing bias or misinterpreting relationships between the respective categories.

Table 6. Injury categories and classification

Injury category	Example
Muscle Injuries	Hamstring Strain, Calf Strain, Groin Pull
Joint Injuries	ACL Tear, Ankle Strain, Meniscus Fracture
Bone Injuries	Leg Fracture, Metatarsal Fracture
Ligament Injuries	Achilles Tear, Patellar Tendon Rupture
Head Injuries	Concussion, Facial Fractures
Trauma Injuries	Rib Bruise, Shoulder Dislocation
Chronic Injuries	Tendonitis, Sports Hernia, Back Pain
Severe Long-Term Injuries	ACL Reconstruction, Hernia Surgery

As shown in Table 7, player roles were simplified into three main categories: Defender, Midfielder, and Attacker. This approach is consistent with previous studies, such as [21], which used similar classifications for position and session type. However, future models may benefit from incorporating more specific roles, such as for instance Wingers or Holding Midfielders, depending on tactical requirements.

Table 7. Player positioning encoding key

Player	Defender	Midfielder	Attacker
Player A	1	0	0
Player B	0	1	0
Player C	0	0	1

2.2.3. Normalizing continuous factors

To ensure consistent scaling, continuous variables such as minutes played per game, passing completion, days between matches, and average fouls received per game were normalized using Min Max Scaling. This method transforms values into a range between 0 and 1, preventing variables with larger scales from disproportionately influencing the simulation. Normalization is especially important when input variables differ widely in magnitude. For example, minutes played can range from 0 to 120, while fouls may only range from 0 to 5. Without normalization, these differences could distort the model's learning process. As noted by [22], this step is indeed essential to maintain fairness in the modeling process and improve the overall reliability of results. The formula used for Min-Max normalization is:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}; \quad (1)$$

where:

- x original value
- $x_{\min}$ minimum value in the dataset
- $x_{\max}$ maximum value in the dataset
- x' scaled value between 0 and 1

For example, a 90-minute appearance can be normalized as follows: $x' = \frac{90-0}{120-0} = 0.75$.

3. Simulation and Expected Outcomes

The simulation framework is designed to estimate the probability of occurrence of injury risk by modeling player workloads under different match conditions and recovery periods. After preprocessing data as it was explained in the previous section, the dataset is going to be input into a Monte Carlo simulation, which will perform multiple iterations to generate individualized injury risk probabilities. The risk levels (Low, Medium, and High) are based on the injury-free play percentages generated from the Monte Carlo simulation. As there is no predefined standard for these ranges, we arbitrarily created them to provide a clear and interpretable way of separating players based on their injury risk.

Players with an injury-free probability of 80% or higher are classified as Low Risk, indicating they are highly likely to remain fit across simulated match scenarios. Those with probabilities between 65% and 79.9% are assigned to the Medium Risk group, reflecting a moderate level of concern and requiring some attention. Players below 65% are labeled High Risk, as they demonstrated a higher chance of injury across simulations.

The discussed injury risk thresholds have been selected to offer a clear and interpretable classification structure in the absence of standardized benchmarks. These thresholds are consistent with common probability bands used in risk communication settings, including health and sports domains, and help support immediate decision-making. Although acknowledged as arbitrary, this framework is intentionally designed to be adaptable. Once simulation results are available, a sensitivity analysis will be conducted to test how variations in threshold cutoffs affect player classifications. This will help ensure the robustness and generalizability of the model under different conditions. Furthermore, the chosen thresholds support practical decision-making related to player rotation and workload planning. For instance, the model may predict that Player A (e.g., Mohamed Salah) has an 85% chance of avoiding injury, categorizing him as Low Risk. In contrast, Player B (e.g., Virgil van Dijk) may have only a 50% chance of remaining injury-free, placing him in the High Risk category. As shown in Table 8, the simulation provides each player's probability of avoiding injury and assigns them to a corresponding risk tier, that is Low, Medium, or High Risk. These classifications reflect expected patterns based on workload and recovery variation, offering actionable insights for optimizing player rotation while reducing injury risk. This approach builds on prior research such as [23], which used Monte Carlo to estimate injury risk from training loads.

Table 8. Expected Monte Carlo simulation outputs

Player	Risk Level	Injury-Free Play (%)
Mohamed Salah	Low	85.2%
Virgil van Dijk	High	50.0%
Sadio Mané	Medium	70.4%
Trent Alexander-Arnold	Medium	78.9%

Table 9. Expected outcomes

Player-based results	Team-based results
Identifying workload thresholds that increase injury risk	Anticipating injury clusters during intense scheduling cycles
Enabling self-assessment for improved training efficiency	Enhancing long-term squad availability through predictive rotation
Highlighting high-risk recovery patterns across individuals	Supporting data-backed decisions in transfer market (risk vs. value)
Allowing players to adapt playstyles reducing recurrence	Helping build customized conditioning strategies per position
Fostering accountability and performance tracking over time	Minimizing financial loss by managing injury-prone players

A comprehensive sensitivity analysis has not yet been formalized. However, it is acknowledged as a critical next step to further assess the robustness of the simulation results as well as the reliability of the proposed model. The current framework is designed to accommodate varying input conditions, including player workload, recovery periods, and contextual match factors, making it well-suited for such evaluation. In future phases, key variables such as minutes played per game, days between matches, and previous injury history will be systematically adjusted to observe their impact on the assigned risk classifications. This process will help identify which inputs exert the most influence on injury likelihood and will contribute to validating the model's interpretability and reliability under dynamic scenarios.

The simulation produces a range of expected outcomes that can support both individual player management and broader team strategy. By estimating the probability of occurrence associated to injury risk under different conditions, the model supports more informed decisions in areas like player rotation, recovery planning, and training load management. For individual players, results help identify suitable risk thresholds, in other terms when someone may be approaching a critical level of risk. This would allow coaches and medical staff to adjust playing time, modify training routines, or implement additional recovery measures to reduce the chance of injury. On a team level, the simulation can reveal broader trends, such as periods when multiple players may be at increased risk. These insights would help managers prepare in advance, ensuring that key players remain available during important stages of the season. Table 9 outlines the actionable benefits of these insights ranging from improved recovery protocols and rotation strategies to informed decisions in transfers and long-term squad development.

To support integration into real-world workflows, the simulation outputs can be incorporated into weekly training reviews, match preparation sessions, and post-match recovery planning. Coaching staff may use player-specific risk tiers to guide rotation decisions or adapt tactical roles based on individual fatigue levels. Similarly, medical teams can prioritize monitoring and treatment resources by flagging high-risk players for preventive assessments or modified training loads. These outputs can be delivered through simple dashboards or reports aligned with existing analysis meetings, making them easily interpretable without disrupting ongoing routines. This integration ensures that simulation insights directly inform practical decisions related to performance management and injury prevention.

While models developed in previous studies focused primarily on internal factors such as player physiology and workload, our work of research expands the existing approach by including such variables such as match conditions, opponent difficulty, and the length of recovery periods between games. By integrating both internal and external factors, this study aims to provide a more comprehensive overview of the complex nature related to injury risk. Both this work and earlier research highlight the significant role of simulation in developing effective injury prevention strategies within professional sports for targeted prevention programs that are both evidence-based and sport-specific.

4. Conclusions and Future Research

The procedure formalized in this paper offers a proactive and evidence-based approach that aligns with the growing role of analytics in modern sports science, providing a foundation for smart decision-making and effective injury prevention strategies. Particularly, the paper outlines a data-driven framework for injury prediction in professional soccer, with a specific focus on thorough data collection, validation, and preprocessing to support simulation-based analysis. As the Monte Carlo simulation approaches completion, this research formalizes the foundational elements of the framework as well as the related expected outcomes, allowing for a smoother transition into the development phases. In this context, the simulation framework is designed to address key limitations in existing injury models by emphasizing important objectives such as flexibility, transparency, and scalability. One of the primary expected outcomes is indeed the ability to adapt the model across different teams, leagues, and even sports. This adaptability can support the management of a wide range of use cases, enabling simulations to be tailored to different match environments, recovery approaches and workload patterns based on the data selected.

Considering the development of future lines of research, the framework can be further enhanced by incorporating advanced techniques such as neural networks. This would allow for real-time injury forecasting and more detailed pattern recognition, making the system more responsive to changing match environments. Furthermore, this approach has potential applications beyond sports and typically in such sectors as healthcare and manufacturing, where injury risk, fatigue or system stress can be modeled and anticipated through similar data-driven strategies. Another important characteristics of this framework is its accessibility. Built with open-source tools and based on publicly available datasets, it is a cost-effective solution that can be used by organizations with limited resources. Future development will focus on completing the simulation module, validating its performance across real-world scenarios, and extending its use with live data from multiple teams and competitions. As part of this process, the framework will be further developed to finalize the Monte Carlo simulation model, perform repeated simulations, and compare the resulting injury risk projections to real-world outcomes. A more formalized sensitivity analysis will also be included in future phases to assess the model's robustness across different input conditions.

References

- [1] Reddy S. (2023) "The Role of Data Analytics in Enhancing Decision-Making in Sports Management." *International Journal of Artificial Intelligence, Data Science, and Machine Learning* 4 (2): 9–16.
- [2] Alamar B. (2024) "Sports analytics: A guide for coaches, managers, and other decision makers." *Columbia University Press*.
- [3] Alghurabi AMK, Rao DS. (2025) "Optimizing Performance with Data Analytics in Sports." *Advances in Sports Science and Technology*: 116–121.
- [4] Chang V, Sajeev S, Xu QA, Tan M, Wang H. (2024) "Football Analytics: Assessing the Correlation between Workload, Injury and Performance of Football Players in the English Premier League." *Applied Sciences* 14 (16): 7217.
- [5] Musat CL, Mereuta C, Nechita A, Tutunaru D, Voipan AE, Voipan D, Mereuta E, Gurau TV, Gurău G, Nechita LC. (2024) "Diagnostic Applications of AI in Sports: A Comprehensive Review of Injury Risk Prediction Methods." *Diagnostics* 14 (22): 2516.
- [6] Robertson S. (2014) "Improving load/injury predictive modelling in sport: The role of data analytics." *Journal of Science and Medicine in Sport* 18: e25–e26.

- [7] Ruddy JD, Cormack SJ, Whiteley R, Williams MD, Timmins RG, Opar DA. (2019) “Modeling the risk of team sport injuries: a narrative review of different statistical approaches.” *Frontiers in Physiology* **10**: 829.
- [8] López-Valenciano A, Ruiz-Pérez I, García-Gómez A, Vera-García FJ, De Ste Croix M, Myer GD, Ayala F. (2020) “Epidemiology of injuries in professional football: a systematic review and meta-analysis.” *British Journal of Sports Medicine* **54** (12): 711–718.
- [9] Rossi A, Pappalardo L, Cintia P, Iaia FM, Fernández J, Medina D. (2018) “Effective injury forecasting in soccer with GPS training data and machine learning.” *PloS One* **13** (7): e0201264.
- [10] Ekegren CL, Gabbe BJ, Finch CF. (2016) “Sports injury surveillance systems: a review of methods and data quality.” *Sports Medicine* **46** (1): 49–65.
- [11] Van Eetvelde H, Mendonça LD, Ley C, Seil R, Tischer T. (2021) “Machine learning methods in sport injury prediction and prevention: a systematic review.” *Journal of Experimental Orthopaedics* **8** (1): 1–15.
- [12] Davis J, Bransen L, Devos L, Jaspers A, Meert W, Robberechts P, Van Haaren J, Van Roy M. (2024) “Methodology and evaluation in sports analytics: challenges, approaches, and lessons learned.” *Machine Learning* **113** (9): 6977–7010.
- [13] Chen Z, Dai X. (2024) “Utilizing AI and IoT technologies for identifying risk factors in sports.” *Heliyon* **10** (11).
- [14] Lahvička J. (2015) “Using Monte Carlo Simulation to Calculate Match Importance: The Case of English Premier League.” *Journal of Sports Economics* **16** (4): 390–409.
- [15] Ekstrand J, Hägglund M, Waldén M. (2011) “Injury incidence and injury patterns in professional football: the UEFA injury study.” *British Journal of Sports Medicine* **45** (7): 553–558.
- [16] Griffin S, Kenny IC, Comyns TM. (2022) “The interval between matches significantly influences injury risk in field hockey players.” *BMJ Open Sport & Exercise Medicine* **8** (1): e001256.
- [17] Zeleny A, Busfield BT, Knesek D, Rosenthal Y, Kaar S. (2021) “Effect of passing plays on injury rates in the National Football League.” *Orthopaedic Journal of Sports Medicine* **9** (11): 23259671211048160.
- [18] Dvořák J, Junge A, Graf-Baumann T, Peterson L. (2000) “Risk factor analysis for injuries in football players: possibilities for a prevention program.” *The American Journal of Sports Medicine* **28** (5): S69–S74.
- [19] Everett G, Beal R, Matthews T, Norman TJ, Ramchurn SD. (2024) “The Strain of Success: A Predictive Model for Injury Risk Mitigation and Team Success in Soccer.” *arXiv preprint arXiv:2402.04898*.
- [20] Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP. (2002) “SMOTE: Synthetic Minority Over-sampling Technique.” *Artificial Intelligence Research* **16**:321–357.
- [21] Freitas T, Ramos R, Ribeiro J, Rebelo-Gonçalves R, Baptista I, Clemente FM. (2025) “Predicting Noncontact Injuries of Professional Football Players Using Machine Learning Approaches.” *Sensors* **24**(2):535.
- [22] Ali PJM, Faraj RH, Koya E. (2014) “Data Normalization and Standardization: A Technical Report.” *Machine Learning Technical Reports* **1**(1):1–6.
- [23] Bornn L, et al. (2019) “Training Schedule Confounds the Relationship between Acute:Chronic Workload Ratio and Injury Risk.” *Journal of Sports Science and Medicine* **18**(4):637–644.